

ALEXANDROS TSAGKAROPOULOS

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EDUCATION

Columbia University Sep 2026 – Present
Ph.D. in Computer Science

Advisors: Associate Professor Asaf Cidon
Assistant Professor Kostis Kaffes

National and Kapodistrian University of Athens Sep 2023 – Mar 2026
M.Sc. in Electronics-Radioelectrology and in Control and Computing

GPA: 9.87/10
Thesis: Evaluation and Performance Optimization of Virtual Memory Support for GPUs
Advisor: Assistant Professor Vasileios Karakostas

National and Kapodistrian University of Athens Sep 2019 – Aug 2023
B.Sc. in Physics

GPA: 9.27/10 (Ranked 1st in class of 2023)
Thesis: Image Segmentation Algorithm for Cloud Recognition from Sky Camera Images
Using Threshold Conditions
Advisor: Professor Dionysios Reisis

EMPLOYMENT

ERMIS: Constellation of Three CubeSats Jul 2024 – July 2026
Embedded Software Engineer

- Implement the control logic of the On Board Computer (OBC) modes of operation.
- Manage the networking of the space segment based on the [CubeSat Space Protocol](#).
- Organize the version control system of the OBC firmware and configuration.
- Integrate the Attitude Determination and Control System (ADCS) library to the OBC SDK.

7L International Sep 2022 – Oct 2022
Software Engineering Intern

- Studied about [React](#) and [Next](#) javascript frameworks.
- Educated the Software Engineering Team about [Qwik](#) and [Remix](#) javascript frameworks.

PUBLICATIONS

Conferences

- [1] Christoforos Vasilakis, Alexandros Tsagkaropoulos, Angelos Motsios, Christos Tsigkanos, and Dionysios Reisis. "[Nanosatellite Flight Software: A Rigorous Software Architecture Perspective](#)." European Conference on Software Architecture (ECSA). September 2025.
- [2] Christoforos Vasilakis, Alexandros Tsagkaropoulos, Angelos Motsios, and Dionysios Reisis. "[Autonomously Reconfigurable Telemetry and Monitoring System for CubeSats](#)." IEEE Computer Society Annual Symposium on VLSI (ISVLSI). July 2025.
- [3] Alexandros Tsagkaropoulos, Christoforos Vasilakis, Georgios Venitourakis, Tzouma Amrou, Georgios Konstantoulakis, Panagiotis Golemis, and Dionysios Reisis. "[Computing the Cleanness of the Photovoltaic \(PV\) Panels](#)." Emerging Tech Conference Edge Intelligence (ETCEI). October 2024.
- [4] Christoforos Vasilakis, Georgios Venitourakis, Alexandros Tsagkaropoulos, Tzouma Amrou, Georgios Konstantoulakis, Panagiotis Golemis, and Dionysios Reisis. "[Comparing the Performance of Cameras and SW Packets for the Cloud Coverage Process in Photovoltaic \(PV\) Parks](#)." Emerging Tech Conference Edge Intelligence (ETCEI). October 2023.

Journals

- [1] Christoforos Vasilakis, Alexandros Tsagkaropoulos, Angelos Motsios, Panagiotis Karathanasis, Christos Tsigkanos, and Dionysios Reisis. "[On Architectural Analysis for Nanosatellite Flight Software: Deconstructing ERMIS3 per 4+1.](#)" Journal of Systems and Software (JSS). August 2026.
- [2] Christoforos Vasilakis, Alexandros Tsagkaropoulos, Ioannis Koutoulas, and Dionysios Reisis. "[Improving the Fast Fourier Transform for Space and Edge Computing Applications with an Efficient In-Place Method.](#)" Software. May 2025.
- [3] Georgios Venitourakis, Christoforos Vasilakis, Alexandros Tsagkaropoulos, Tzouma Amrou, Georgios Konstantoulakis, Panagiotis Golemis, and Dionysios Reisis. "[Neural Network-Based Solar Irradiance Forecast for Edge Computing Devices.](#)" Information. November 2023.

Posters & Workshops

- [1] Christoforos Vasilakis, Alexandros Tsagkaropoulos, Angelos Motsios, Panagiotis Karathanasis, Christos Tsigkanos, and Dionysios Reisis. "[Reducing Onboard Power Consumption in CubeSats via Adaptive Telemetry and Monitoring.](#)" (ACSOS-C). September 2025.

AWARDS

- State Scholarships Foundations (IKY) Award of Excellence Apr 2024

TEACHING EXPERIENCE

Lab Volunteering: "Algorithms and Data Structures" Fall 2025 – 2023

- Introduction to C programming language.
- Implementation of data structures (Linked List, BST, RBT) and algorithms (Merge/Heap Sort).
- Detection of cycles in a directed graph with Breadth First Search.

Mentorship

- Anastasia Zamani. "Image Classification of Waste Data Using Convolutional Neural Networks." Bachelor's Thesis. May 2026.
- Vasileios Zachos. "The Tower of Light." Bachelor's Thesis. March 2026.
- Aikaterini Tsapraili. "Development of Neural Networks for Handwritten Digit Recognition Using the MNIST Dataset." Bachelor's Thesis. August 2025.
- Nikolaos Chatzoglou. "Image Modality Control Algorithm and Cloud Recognition Using Threshold Conditions from a Sky Camera." Bachelor's Thesis. October 2024.

PROJECTS

Tower of Light ([link](#)) Aug 2024 – Sept 2024

- Implemented a rogue-like RPG game in Java.
- Utilized Composite, Decorator and Observer patterns for the UI.

ARM Multicycle Processor ([link](#)) Feb 2024 – Apr 2024

- Implemented a subset of ARMv4 ISA using the Harvard architecture.
- Developed automatic generation scripts for the Instruction Decoder and the Stage FSM.

ATLAS Experiment: Displaced Vertex Identification ([link](#)) Nov 2021 – Oct 2022

- Formalized the algorithm that constructs Long-Lived Particles from its product's trajectories.
- Surpassed the average human input gathered from the [zooniverse.org](#) platform.

Graphics with TikZ ([link](#)) Sep 2021 – Present

- Create graphics for publications, presentations, projects and lab reviews.
- Contributed the source code to [tikz.net](#).